Social Network Structures and Information Flow Processes: Implications for Estimating Epidemics

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Abstract

Social media data, particularly Twitter, have been proposed as potential surveillance tools in epidemiology. However, previous research demonstrates that Twitter data alone is an insufficient proxy for population-level data in the development of predictive models of disease transmission. Deviations in disease prevalence and incidence estimates between Twitter and actual surveillance data may be due in part to the network structure and to the processes of information dissemination between users in the social media population. We reconstructed hypothetical contact networks using a sample of Twitter users to study how the structure of their networks results in changes to simulated disease reporting incidence as compared to actual Twitter network post data. By increasing our understanding of the flow of information dissemination through social media channels, we may be able to improve existing methods of disease surveillance and transmission estimation through social media resources.

Keywords: Twitter, network analysis, disease modeling, network structures, social media

Introduction

Social media has recently become a popular tool to track and predict disease epidemics.[3,4] Previous studies have proposed that social media streams on platforms such as Twitter can contribute to US Center for Disease Control (CDC) surveillance systems given their large and diverse user bases, and real-time features.[1-3] However, few studies have investigated how network structures in social media and their user behaviors combine and interact to affect micro-blogging trends in the context of epidemic monitoring.

Methods

We model two types of network structures using Twitter data 1) A Twitter network from a confined geographical region, and 2) A configuration model based on the degree distribution of the same Twitter network. We then implement a compartmental epidemical model (SIR) to simulate flu epidemics through each of the networks, and run parameter sweeps on reporting behavior probabilities of users. We compute the correlation between actual stream data and simulated results in order to investigate the difference between our simulated posts flu reporting data and actual flu reporting streams on Twitter. By comparing our model outcomes to the actual social media disease-specific post stream content, we reveal the potential impacts of network structure and user's reporting behavior during disease epidemics, as well as possible inferences between social media networks and their corresponding contact networks.

References

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